# Earnings Call Transcript Analysis

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# Problem Statement

## The Question

Do executives' words during earnings calls contain predictable signals about short-term stock price direction?

## The Hypothesis

NLP-derived features from transcript text can outperform a random baseline in predicting both the direction and magnitude of post-earnings stock moves.

# Dataset Overview

## Dataset 1 — Motley Fool Transcripts

- Scraped quarterly earnings calls from Motley Fool
- Fields: ticker symbol, earnings date, full transcript text
- Covers multiple quarters across large-cap US stocks
- 500 transcripts sampled for this project
- Transcripts average ~8,000–10,000 words each

## Dataset 2 — yfinance Augmentation

- Yahoo Finance historical OHLCV data via yfinance API
- Price before: closing price on or before earnings date
- Price after: closing price 3 trading days post-earnings
- Binary target: 1 = price up, 0 = price down
- Continuous target: % change between before and after

# Exploratory Data Analysis

**207 / 176**

Down vs Up splits

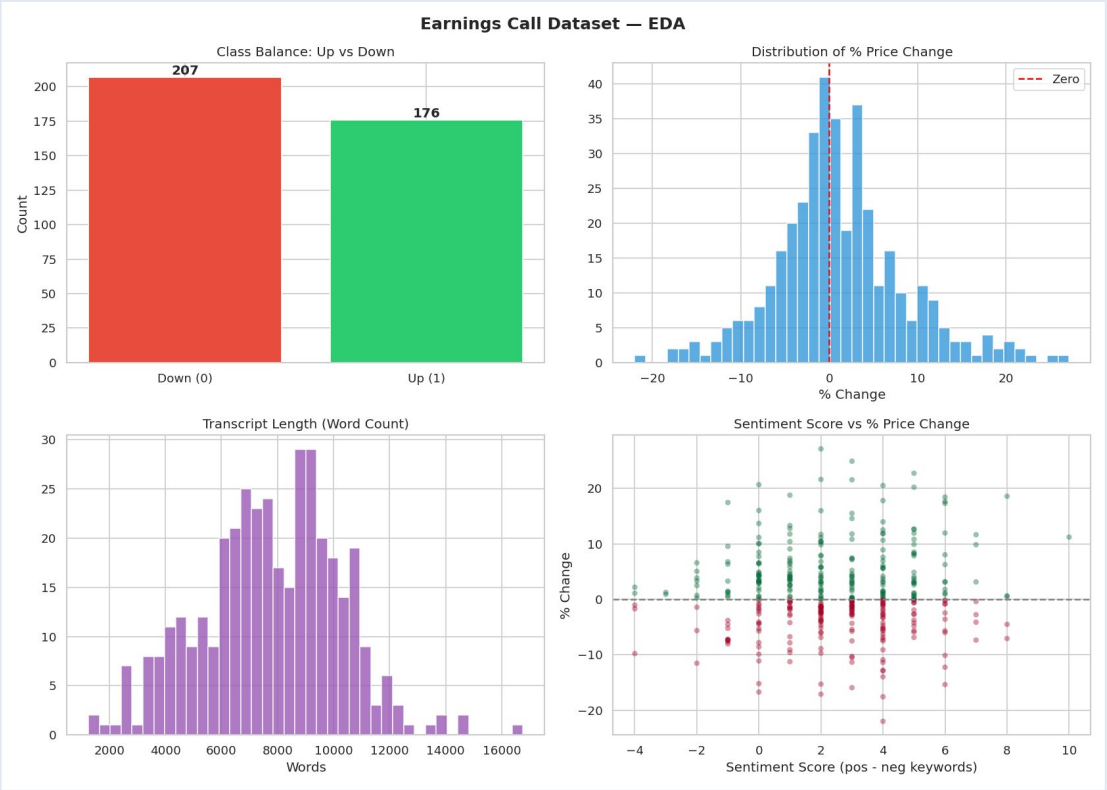
**±6.5%**

Typical % move range

**~8,500**

Avg transcript words

**Mild class imbalance (54%/46%). % change distribution is approximately normal.**



# Preprocessing & Feature Engineering

**01**

## Text Cleaning

- Lowercase all text
- Strip URLs, brackets, punctuation
- Collapse whitespace
- Remove transcripts < 50 words

**02**

## TF-IDF Vectorization

- 5,000 features, unigrams + bigrams
- Sublinear TF scaling
- Min/max document frequency filters
- Sparse matrix representation

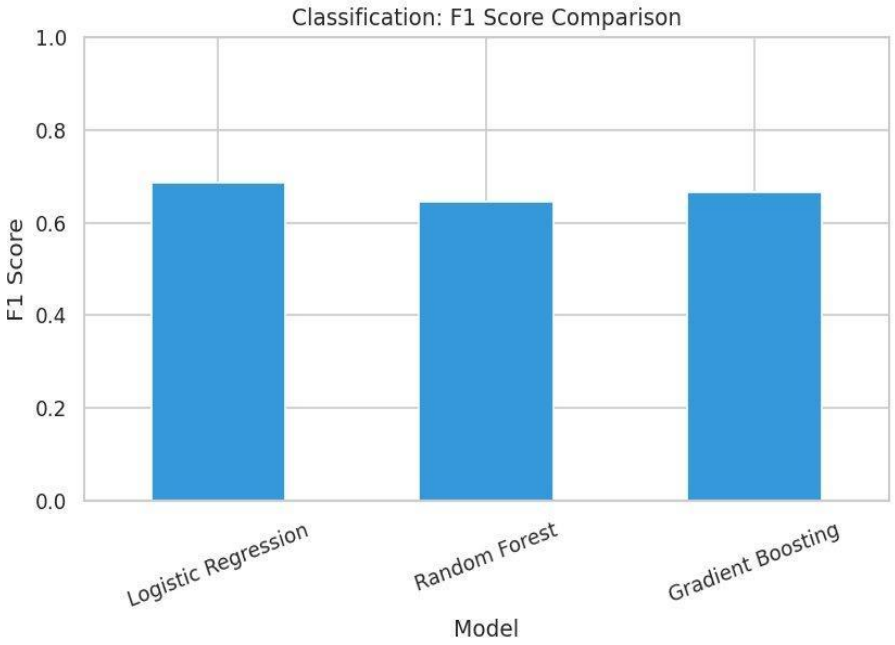
**03**

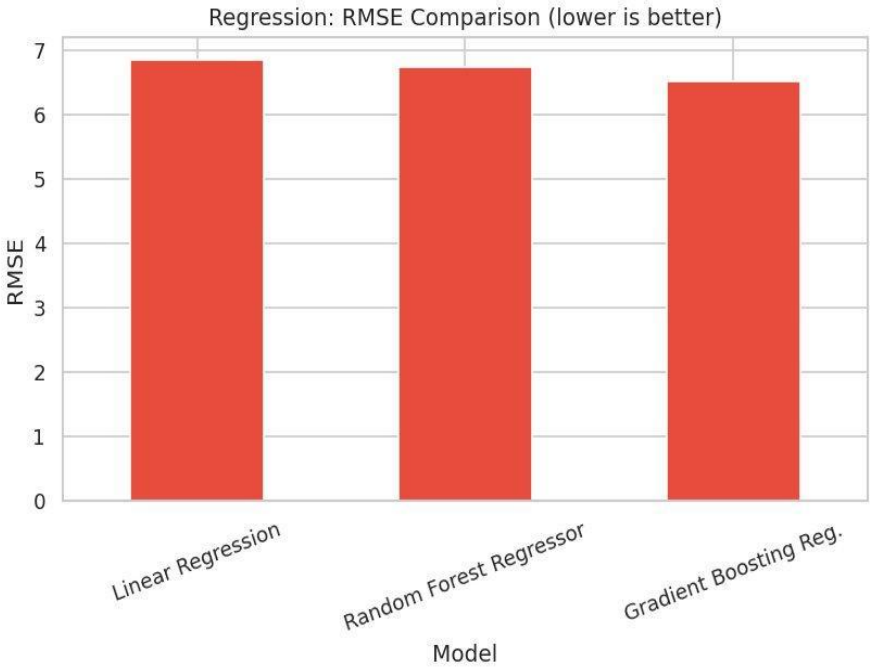
## Structured Features

- Transcript word count
- Positive keyword count (12 words)
- Negative keyword count (12 words)
- Net sentiment score (pos − neg)

*Final feature matrix: 5,000 TF-IDF + 4 structured features = 5,004 dimensions per transcript*

# Model Selection



# Model Comparison Summary

| Model | Task | Key Metric | Result |
|---|---|---|---|
| Logistic Regression | Classification | F1 | 0.69 |
| Random Forest | Classification | F1 | 0.64 |
| Gradient Boosting | Classification | F1 | 0.67 |
| Linear Regression | Regression | RMSE | 6.9 |
| Gradient Boosting Reg. | Regression | RMSE | 6.6 |

# Hyperparameter Tuning

RandomizedSearchCV with 3-fold CV, 15 iterations each model

## Random Forest Classifier

### Parameter Grid

| | |
|---|---|
| **n_estimators** | 100 – 300 |
| **max_depth** | None, 10, 20, 30 |
| **min_samples_split** | 2, 5, 10 |
| **max_features** | sqrt, log2 |

Scored on F1

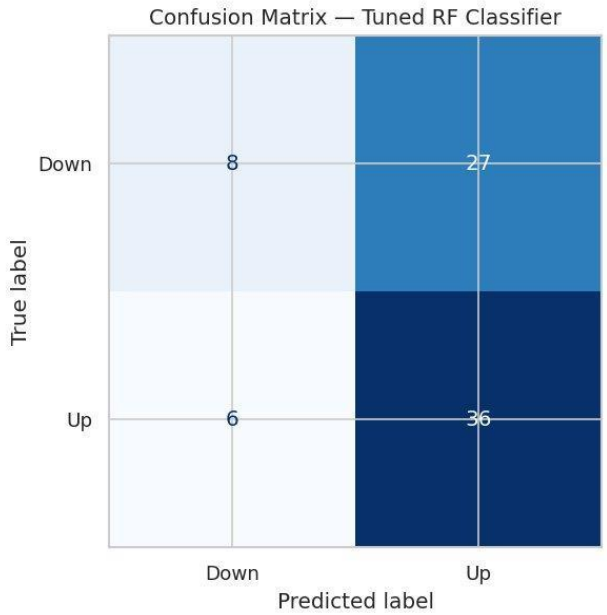
## Gradient Boosting Regressor

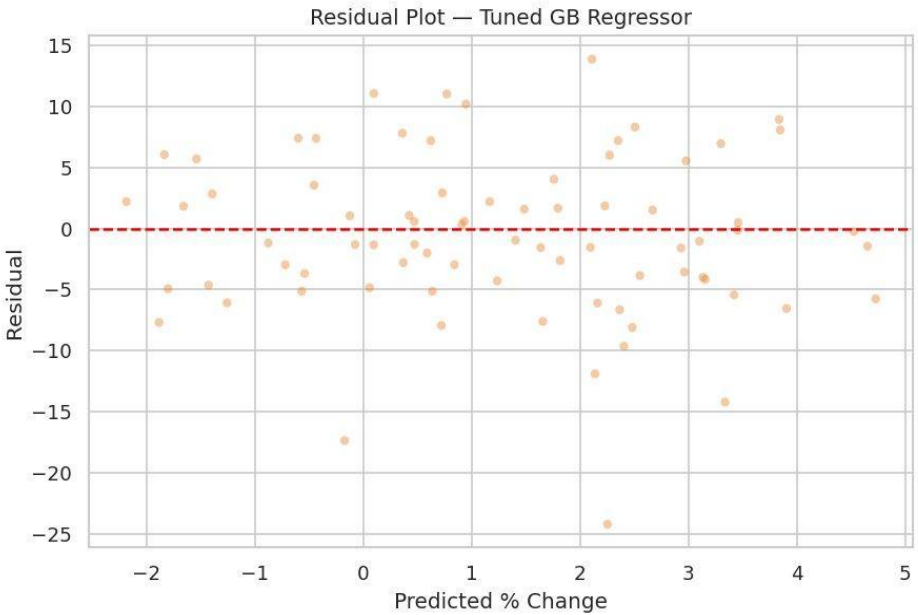
### Parameter Grid

| | |
|---|---|
| **n_estimators** | 100 – 300 |
| **learning_rate** | 0.05, 0.1, 0.2 |
| **max_depth** | 3, 5, 7 |
| **subsample** | 0.7, 0.8, 1.0 |

Scored on neg-RMSE

# Final Model Evaluation




## Confusion Matrix

RF Classifier predicts UP heavily (36 correct, 6 false negatives). Strong recall for up-moves; weaker at catching real down-moves. Reflects mild class imbalance.

## Residual Plot

Residuals scatter randomly around zero indicating no systematic bias. Spread reflects the inherent unpredictability of short-term price changes from text alone.

# Results at a Glance

## Classification — Tuned RF

**Accuracy** **~57%**
Above random (50%)

**Precision** **~58%**
Of predicted up, 58% correct

**Recall** **~86%**
Finds most true up-moves

**F1 Score** **~0.69**
Best CV score after tuning

## Regression — Tuned GB

**RMSE** **~6.6%**
Avg error in percentage points

**MAE** **~4.5%**
Mean absolute error

**$R^2$** **~0.04**
Low — text alone is insufficient

**Trend** **Unbiased**
Residuals centered at zero

# Limitations & Challenges

### Weak Signal from Text

Earnings calls are carefully scripted. Executives rarely reveal material information that hasn't already been priced in, creating a fundamental gap between language and price movement.

### Market Randomness

Short-term stock movements are driven by macro factors, sector momentum, and market microstructure — forces entirely outside the transcript. This explains the low regression $R^2$.

### Data Merging Gaps

yfinance returns no data for tickers on trading holidays, suspended stocks, or delisted companies, reducing the usable dataset and potentially introducing survivorship bias.

### No Temporal Split

A random train-test split may allow look-ahead bias. A chronological split (train on older calls, test on newer) is more realistic but reduces training data size for smaller datasets.

# Future Improvements

## 1 FinBERT Embeddings

Replace TF-IDF with domain-specific pre-trained transformer embeddings (FinBERT) trained on financial text for richer semantic representations.

## 2 Chronological Split

Train on calls before 2022, validate on 2022–23, test on 2024+. This eliminates look-ahead bias and tests real-world generalization.

## 3 Q&A vs. Remarks

Separate the prepared executive remarks from the analyst Q&A section. The Q&A may contain more genuine signals as executives respond to probing questions.

## 4 Macro Features

Add sector, market cap, analyst consensus estimate vs. actual EPS/revenue as structured features to give models richer context beyond text alone.

# Conclusion

### Classification: Above Chance

Random Forest F1 ~0.69. Transcript text does carry directional signal, though it is modest. High recall for UP moves is actionable.

### Regression: Hard Problem

RMSE ~6.6% with near-zero $R^2$ confirms that predicting the magnitude of price moves from text alone is unreliable. Structural market data is needed.

### Key Takeaway

Earnings calls provide weak but real NLP signals for direction. This project demonstrates a complete, reproducible pipeline to create sophisticated models.